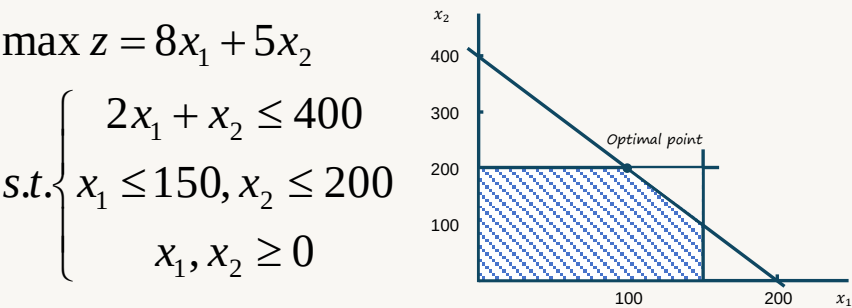


Mathematical Programming (MP)

Produce two hat models using **limited daily labor**. Model 1 needs **2x labor** of Model 2. **Max daily capacity is 400** (if only Model 2). Demand limits: **150** (Model 1) and **200** (Model 2). Selling prices are **\$8** and **\$5**. Determine optimal quantities to **maximize total revenue**.

Linear Programming Formulation



Dynamic Optimization (DO)

A factory manages **100 machines** over **four cycles**. Each cycle, some machines are assigned to **Task 1 (Revenue 10)**, where **1/3 fail**. The survivors join the unassigned machines to perform **Task 2 (Revenue 7)**, after which **1/10 fail**. Machines **surviving both tasks are available for the next cycle**. Determine the number of machines to assign to Task 1 in each cycle to **maximize total cumulative revenue**.

Discrete-Time Dynamic Programming Model

State Transition : $S_{t+1} = 0.9(S_t - \frac{x_t}{3})$

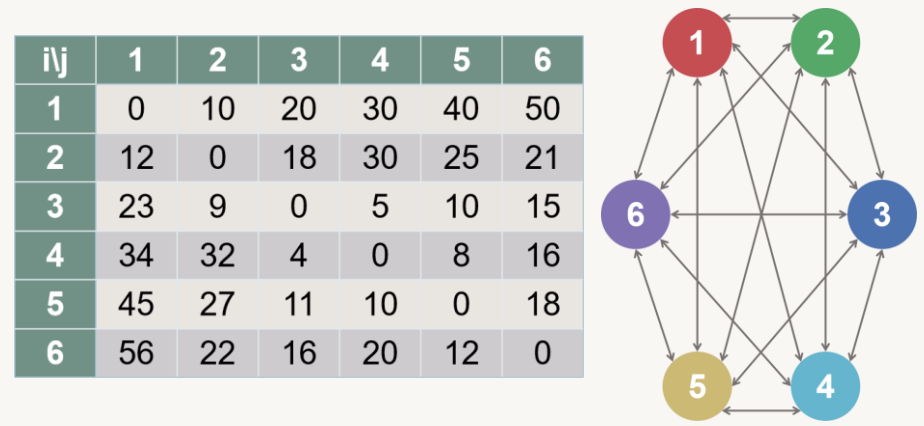
Bellman Equation : $V_t(S) = \max_{0 \leq x \leq S} \{10x_t + 7(S_t - \frac{x_t}{3}) + V_{t+1}(S_{t+1})\}$

Boundary Conditions : $V_5(S) = 0 \quad \forall S \geq 0$

Combinatorial Optimization (CO)

A traveling salesman **starts at City 1**, visits 5 other cities exactly once, and returns to City 1. Given the **pairwise distance matrix** for these 6 cities, determine the visiting order that **minimizes the total travel distance**.

Distance Matrix for Tour Optimization



Optimal Control (OC)

A lunar module must achieve a **soft landing (zero altitude and velocity)**. **Thrust can be adjusted** to affect motion and fuel consumption. Given initial conditions and dynamics, determine the **thrust policy over time that minimizes total fuel usage** while satisfying **landing requirements**.

System Dynamics & Constraints

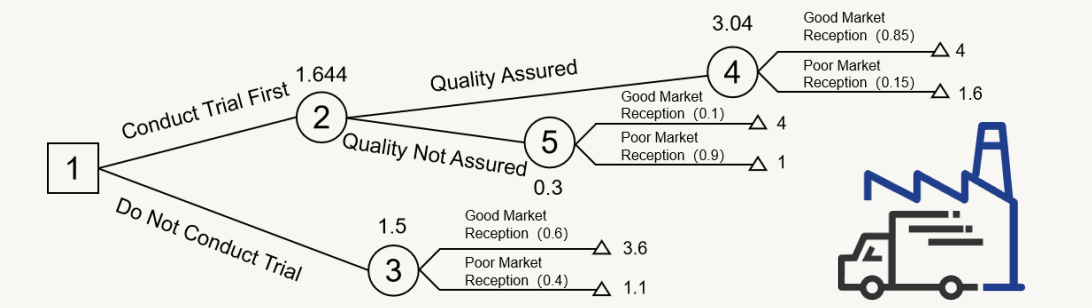
Dynamics Model :
$$\begin{cases} \dot{h}(t) = v(t) \\ \dot{v}(t) = \frac{u(t)}{m(t)} - g \\ \dot{m}(t) = -ku(t) \end{cases}$$

Boundary Conditions : $h(t_f) = 0, v(t_f) = 0$

Stochastic Optimization (SO)

A factory considers a trial test (**cost 0.3**) for a new product. With trial, quality is **assured (60%)** or **not (40%)**. Assured quality has **cost 0.6** and outcomes: Good (**prob 0.85, rev 4**) or Poor (**prob 0.15, rev 1.6**). Unassured quality has **cost 1.0** and outcomes: Good (**prob 0.10, rev 4**) or Poor (**prob 0.90, rev 1.0**). **Without trial**, production **cost 1.1** yields outcomes: Good (**prob 0.60, rev 3.6**) or Poor (**prob 0.40, rev 1.1**). Determine the decision that **maximizes expected profit**.

Stochastic Decision Tree under Uncertainty



Game Optimization (GO)

Clinic A (strategies α_1, α_2) competes with Clinic B (strategies β_1, β_2) quarterly. The net gains/losses are: with (α_1, β_1) , **A loses 5 and B gains 5**; with (α_1, β_2) , **A gains 10 and B loses 10**; with (α_2, β_1) , **A gains 5 and B loses 5**; with (α_2, β_2) , **both break even at 0**. Assuming indefinite repetition, determine the optimal long-term strategy Clinic A should adopt to **maximize its total financial advantage**.

Bimatrix Game Payoff Structure and Strategies

	Outreach (β_1)	Discount (β_2)
Specialist (α_1)	(-5, 5)	(10, -10)
Hours (α_2)	(5, -5)	(0, 0)